Final Exam: Topics

Material covered

- Lectures 16-21, Labs 7-8, AE 5-6

Coding

- `ggcorrplot()`, `cor()`, `pcor.test()`
- `ggpairs()`
- How to subset columns/rows using brackets: i.e. `iris[,1:4]`
- `facet_wrap()`
- `+ geom_smooth(method = "lm")`
- `tidy()` and `kable()`
- `vif()`
- What does "tidy format" mean for a dataset?
- `glance()` and `augment()`
- What does the double colon specify? (e.g. `dplyr::select()`)
- `lm()`. How to fit a linear regression model?
- `glm()`. How to fit a logistic regression model?
- `predict()`
- Differences in code for how to fit and predict outcomes for a linear vs. logistic regression model

Conceptual and Applied

- Knowing common tests or models given the type of predictor and type of outcome ('When to choose what method' at the beginning of Lec 21).
- Is correlation between X with Y the same as correlation between Y and X? What about the linear regression of Y on X versus the linear regression of X on Y?
- Correlation coefficient ρ: what type of relationship does it describe? Know the range and how to interpret positive/negative/zero values. Which statistic can we use to estimate ρ? (How to do this in `R`?)

    - Spearman vs. Pearson
- Partial correlation. What is being tested?
- What is the takeaway of Anscombe's quartet?
- Confounding definition
- What is a regression line? What are residuals? What is error?
- Know how to write out a linear model for a particular study, indexed by observation $i = 1, \ldots, n$.Be able to define all terms in the model.
- Interpreting coefficients from simple and multiple linear regression models. What's the difference between simple vs. multiple linear regression?
- Model assumptions of linear regression: be able to describe, and also how to check each one in R
- In regression output, what are the individual coefficient statistics and p-values telling us? Which hypothesis test is being conducted for each p-value? Be able to write out the null and alternative hypotheses in symbols and in words.
    - For the global F-test, how would you interpret a p-value from the overall test? Be able to write out the null and alternative hypotheses in symbols and in words.
- R^2 and Adjusted R^2: definitions and how to interpret. What's the difference between them?
- How to predict a new value based on estimated coefficients from a regression model? (e.g. writing out an expression)
- How to construct a confidence interval for $\hat{\beta}_k$?
- How does R treat categorical variables in a regression model? Know how to interpret a dummy variable from a regression model.
- Interaction models: How to fit in R? How to interpret a main effect when an interaction effect is present in the model?
- Multicollinearity: definition & how to detect (e.g. VIF)
- How does logistic regression differ from linear regression?
- In logistic regression, how do you interpret the estimated coefficients in terms of odds? How do you predict log-odds for a new observation given the estimated model coefficients?
- How do you interpret the p-value for a single interaction term (like examples from Lecture 19)
- How to predict a new value based on estimated coefficients from a logistic regression model? (e.g. writing out an expression) What does this value represent for this individual?